

Secure Application Model Notes



Type:	7. Reference	Version:	1.0
Status:	Final	Effective:	7/21/2025

1. Purpose

Access to Partner Center requires MFA; however, service principals are non-interactive so they cannot perform MFA.

Microsoft supports the Secure Application Model framework which uses a user account to perform MFA then links it to the service principal.

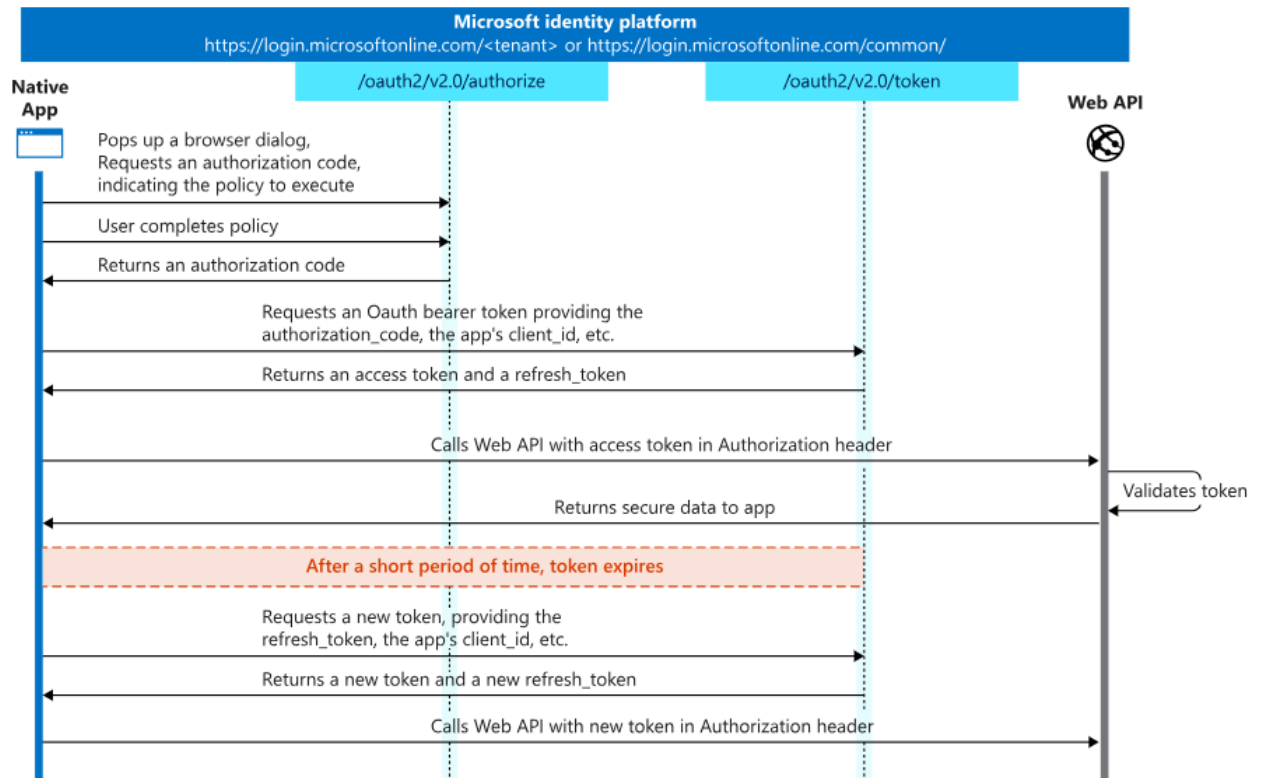
TD SYNEX does not offer programming assistance. This document is not intended to provide guidance on how to use the Secure Application Model. Use if this document is at the user's own risk.

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3. Notes

3.1. OAuth 2.0 Authorization Code Flow



3.2. Authorization Code Endpoints

Ensure that the correct endpoints are used.

Cloud	Endpoint
Commercial	<code>https://login.microsoftonline.com/common/oauth2/authorize</code>
US Government	<code>https://login.microsoftonline.us/common/oauth2/authorize</code>

3.3. Request Authorization Code

Example:

`https://login.microsoftonline.com/common/oauth2/authorize?client_id={{applicationId}}&response_type=code&response_mode=query&scope=offline_access+openid&redirect_uri=http%3A%2F%2Flocalhost%2Fmyapp`

Note:

- The end point must be for the correct cloud - .com or .us

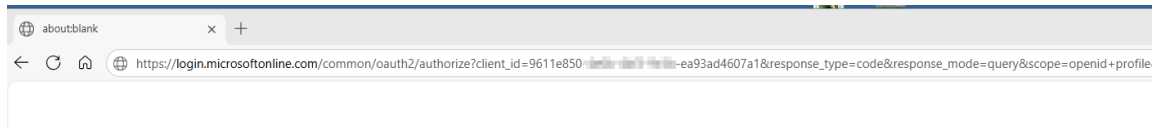
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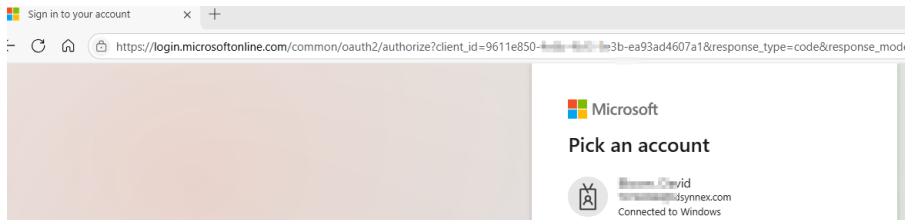
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- The client_id is the client id from the Application Registration
- The scope may change depending on what is needed. The example should work for most authorization requests unless less permissions are needed.
- You must request offline_access to get a refresh token.
- The redirect_url must be one of the [redirect URLs](#) authorized in the Application Registration.

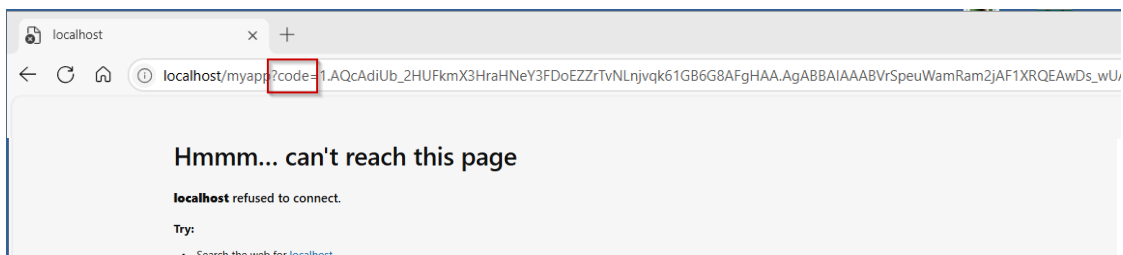
1. Create the request URL similar to the example above.
2. Paste the URL into your web browser and execute



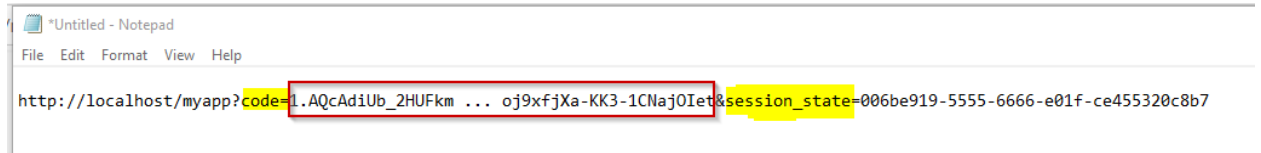
3. When prompted, login with an appropriate user credential.



4. The authorization code will be provided on the URL bar. Copy the entire code to a Notepad. Ignore the "Can't reach this page" notice if received.



5. The authorization code will be a long string of characters after "code=" and before "session_state="



Note: The code was significantly shortened in this example.

3.4. Trade Authorization Code for a Refresh Token

An authorization code does not provide access to resources. It must be traded for an access token to gain access to resources; however, access tokens have a short lifespan. Request a refresh token then use the refresh token to request an access token when needed.

Example:

<https://login.microsoftonline.com/{{tenantId}}/oauth2/v2.0/token>

Key	Value	Description
☒ client_id	{{client_id}}	Partner Center application (client) id
☒ grant_type	authorization_code	We have an authorization code
☒ code	{{authorizationCode}}	The authorization code we have
☒ redirect_uri	http://localhost/myapp	Where to redirect with the refresh token
☒ client_secret	{{client_secret}}	Partner Center client secret for the application (client) id
☒ scope	offline_access openid	

1. In Postman, enter the required information and include the authorization code retrieved in the previous step.

Key	Value	Description
☒ client_id	{{client_id}}	Partner Center application (client) id
☒ grant_type	authorization_code	We have an authorization code
☒ code	{{authorizationCode}}	The authorization code we have
☒ redirect_uri	http://localhost/myapp	Where to redirect with the refresh token
☒ client_secret	{{client_secret}}	Partner Center client secret for the application (client) id
☒ scope	offline_access openid	

Note: The scope must include `offline_access`.

2. Send the request.
3. You will receive an access token and a refresh token.

```
{
  "token_type": "Bearer",
  "scope": "profile openid email Directory.AccessAsUser.All User.Read UserAuthenticationMethod.Read UserAuthenticationMethod.Read.All",
  "expires_in": 4285,
  "ext_expires_in": 4285,
  "access_token": "eyJ0eXAiOiJKV1QiLCJ...nEJTxCCA",
  "refresh_token": "1.AVoAdiUb_2HUFkmX...PhaFUmjQM",
  "id_token": "eyJ0eXAiOiJKV1QiLCJ...0rUsqk68I"
}
```

Note: The tokens were significantly shortened in this example.

4. Save the refresh token in a safe place.

3.5. Get Access Token from Refresh Token

When access to a resource is required, exchange the refresh token for an access token.

When an access token is returned, a new refresh token will also be returned. Save the new refresh token in place of the old one. The new refresh token will have a new expiration date; therefore, if you get a new one on a regular basis, you will never have to do the authorization code again.

Example:

https://login.microsoftonline.com/{{tenant_id}}/oauth2/v2.0/token

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Key	Value	Description
☒ client_id	{{client_id}}	Partner Center application (client) id
☒ grant_type	refresh_token	We have a refresh token
☒ refresh_token	{{refresh_token}}	The refresh token we have
☒ redirect_uri	http://localhost/myapp	Where to redirect with the refresh token
☒ client_secret	{{client_secret}}	Partner Center client secret for the applicaiton (client) id

1. In Postman, enter the required information and include the authorization code retrieved in the previous step.

Key	Value	Description
☒ client_id	{{client_id}}	Partner Center application (client) id
☒ grant_type	refresh_token	We have a refresh token
☒ refresh_token	{{refresh_token}}	The refresh token we have
☒ redirect_uri	http://localhost/myapp	Where to redirect with the refresh token
☒ client_secret	{{client_secret}}	Partner Center client secret for the applicaiton (client) id

Note: The scope must include offline_access.

2. Send the request.
3. You will receive an access token and a refresh token.

```
{
  "token_type": "Bearer",
  "scope": "profile openid email Directory.AccessAsUser.All User.Read UserAuthenticationMethod.Read UserAuthenticationMethod.Read.All",
  "expires_in": 3789,
  "ext_expires_in": 3789,
  "access_token": "-gwroM30VHHQ5WZ453KT788bFPu...DxY2yWdQ7_RTNjPgvSw_xA",
  "refresh_token": "1.AVoAdiUb_2HUFkmX3HraHNeY...bkNs3fI3RlhSy9R3VtPkxg",
  "id_token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzIuNiZy99133pludK25Hm8uiw"
}
```

Note: The tokens were significantly shortened in this example.

4. Use the access token in your API request.
5. Save the refresh token in a safe place – replacing the prior one.

4. Associated Documents

- [Microsoft identity platform and OAuth 2.0 authorization code flow - Microsoft identity platform | Microsoft Learn](#)
- [Enable secure application model - Partner app developer | Microsoft Learn](#)
- [Overview of permissions and consent in the Microsoft identity platform - Microsoft identity platform | Microsoft Learn](#)
- [How to register an app in Microsoft Entra ID - Microsoft identity platform | Microsoft Learn](#)

5. Document Approval

This document has been approved by:

- Manager, Cloud Sales Engineering

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6. Document Revision History

<i>Number</i>	<i>Effective Date</i>	<i>Revision History</i>	<i>Author</i>
1.0	7/21/2025	Initial Draft	David Bloom